Management of *Clostridioides difficile* infection in patients with haematological malignancies and after cellular therapy: guidelines from 10th European Conference on Infections in Leukaemia (ECIL-10)

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Summary

Clostridioides difficile infection (CDI) poses a significant challenge in patients with haematological malignancies (HM) and those undergoing cellular therapy such as haematopoietic cell transplantation (HCT) or CAR T-cell therapy. These patients have high rates of both colonization with _Clostridioides difficile_ and diarrhoea due to non-infectious causes, leading to challenges with establishing diagnosis and optimal management of CDI, especially in the setting of molecular detection of toxin genes alone. Current severity criteria are of limited usefulness since underlying haematological disease and its treatment impact white blood count and inflammatory manifestations of severe CDI. Extensive exposure to antibiotics, profound microbiota damage and bidirectional relationship with gastro-intestinal graft-versus-host disease after transplant further complicate clinical management. Therefore, the 10th European Conference on Infections in Leukemia (ECIL-10) group comprehensively reviewed the literature (published 01/01/2010–15/09/2024) on the epidemiology, treatment and prevention of CDI, and formulated consensus recommendations for the management of CDI specific to this population. New definitions of proven, probable and possible CDI in this population were developed and proposed for use in clinical research to standardise reporting.

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Introduction

Clostridioides difficile (*C. difficile*) is an anaerobic, gram-positive, spore-forming bacillus, with strains that can carry or not toxin genes (toxigenic and non-toxigenic). Colonization is rare in healthy adults, but significantly more common in hospitalized patients (reaching 25%). In adult haematopoietic cell transplant (HCT) recipients and children with haematological malignancies (HM) or HCT, asymptomatic colonization rates of up to 39% and 70%, respectively, have been reported.[1]

C. difficile can either colonize the gastrointestinal tract without causing symptoms or lead to *C. difficile* infection (CDI). CDI is a toxin-mediated disease where two exotoxins, enterotoxin A (TcdA) and cytotoxin B

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Research in context

Evidence before this study

A Pubmed search was conducted for published literature on *Clostridioides difficile* infection in patients with haematological malignancies (HM) and after haematopoietic cell transplantation (HCT) or Chimeric Antigen Receptor T-cell (CAR-T) therapy between 1st January 2010 and 15th September 2024 utilising the search terms ((*C. difficile*) AND (HCT) OR (haematological malignancy) OR (neutropenia) OR (leukemia) OR (lymphoma)); in addition randomised controlled trials (RCTs) and meta-analyses on CDI treatment or prevention in the general population were reviewed, with focus on data in HM/HCT patients, if available.

Despite being at high risk for CDI and its recurrence, there were few specific guidelines on CDI management in HM and HCT patients, utility of severity criteria used for the general population was limited and there was a lack of standardised definitions of CDI used across studies for this target population.

Added value of this study

The European Conference for Infections in Leukaemia (ECIL) sought through this review to critically evaluate the literature on the epidemiology of CDI, diagnostic methods, severity criteria, definitions used as well as clinical treatment trials with a specific focus on data relevant to patients with HM and HCT.

After summarizing available relevant data from observational studies and RCTs, recommendations on diagnosis, treatment, and prevention were developed, graded according to published criteria, discussed by all participants to achieve consensus and were opened for public consultation. Aside from specific recommendations for CDI management, new severity criteria and definitions for proven, probable and possible CDI for clinical research were proposed for this patient population.

Implications of all the available evidence

These guidelines serve as a useful resource for clinical care and will drive improvements in the management of CDI from diagnosis to treatment and prevention.

However, they also illustrate the need for more studies and trials specific to the HM and HCT populations utilising standardised definitions. The proposed definitions of severity and of CDI for use in clinical research for this population is a key step to advancing research.

(TcdB), are responsible for diarrhoea and inflammation. The severity of CDI can vary widely, from mild diarrhoea to severe conditions like pseudomembranous colitis, ileus, or toxic megacolon. Hypervirulent strains, such as NAP1/027 or 078, which produce a binary toxin, have been reported to cause outbreaks of severe colitis, but have not been consistently implicated in HCT settings.[2] The most important predisposing factor for CDI is the disruption of gut microbiota, often induced by antibiotics, which also contributes to high rates of recurrent CDI (rCDI), exceeding 30% in some immunocompromised patients.[3–6] Other predisposing factors common in HM are prolonged hospital stays, immunosuppressive therapies, and mucositis.[7]

The diagnosis of CDI in patients with HM or undergoing HCT remains challenging. Enzyme immunoassays (EIAs) for toxin detection, while specific and relatively low-cost, lack sufficient sensitivity (57%), often yielding false negative results due to toxin instability.[8–10] Reference methods such as Cell Cytotoxicity Neutralization Assay (CCNA), are not suitable for rapid diagnosis.[9,11] On the contrary, toxin gene detection via nucleic acid amplification tests (NAATs) is highly sensitive but unable to differentiate between colonization and active CDI, possibly leading to overdiagnosis and overtreatment.[12] Indeed, in a population with high rate of colonisation and frequent non-infectious diarrhoea due to chemotherapy or graft-versus-host disease (GvHD), diagnosing a true episode of CDI and evaluating treatment response is difficult.[13–15] The negative consequences of overtreatment include delay in treatment of other potential causes of diarrhoea, selection of resistant bacteria (e.g. vancomycin-resistant enterococci), and further microbiome disruption, which has been associated with increased risk of GvHD and lower survival.[16]

Challenges in CDI management in HM/HCT setting also relate to the differences in severe CDI rates due to immunosuppression and the lack of specific criteria to define severity and high risk of poor outcomes. In addition, few therapeutic trials included HM/HCT patients and management strategies such as length of treatment could be different from the general population.

In this setting, the aim of the 10th European Conference on Infections in Leukemia (ECIL-10) guidelines was to review the available data on *C. difficile* in HM and cellular therapy population and provide recommendations on management of CDI in this patients' setting in order to improve diagnostic, therapeutic and prophylactic approach, patients's outcomes and clinical research.

Methods

Guideline development overview

European Conference on Infections in Leukemia (ECIL) is a society co-founded by the Infectious Diseases Working Party of the European Society for Blood and Marrow Transplantation (IDWP of EBMT), the International Immunocompromised Host Society

(ICHS), the European Leukemia Net (ELN), and the European Organisation for Research and Treatment of Cancer (EORTC). The objective of the ECIL is to develop evidence-based guidelines for the management of infectious complications in patients with HM or undergoing HCT, and guideline development methodology has been previously described.[17] In summary, the ECIL Organization Committee selects the topics to be addressed, appoints a designated leader to each one, and invites to the meeting representatives of the affiliated organizations and other international experts, including haematologists, transplantation experts, infectious disease specialists, and microbiologists with knowledge and publications on selected topics in leukemia and HCT populations. Junior members with expertise on the selected topics are also invited (1–2 per topic). The final selection should be balanced with regard to country of origin, gender, and specialty (hematologist and infectious diseases specialists). Group leaders select the panels dedicated to specific topics based on the same criteria. Updates of ECIL guidelines are performed based on new developments in the specific field, and decided by ECIL Council and Organization Committee every 2 years.

All members of the panel dedicated to *Clostridioides difficile* topic performed literature review and elaborated the recommendations graded according to the European Society of Clinical Microbiology and Infectious Diseases (ESCMID) (Table 1), first online and at the 10th ECIL conference (Sept 19–21, 2024, Nice, France). All 46 ECIL-10 participants received the literature analysis and proposed recommendations ahead of the meeting. During the meeting the proposed guidelines were discussed and revised until a consensus, which was defined as simple majority (>50%), was reached, and all group members found the resolution acceptable. The final version was approved on 21/09/2024, and the slide set was made available on the ECIL website on 06/10/2024 for public consultation for external feedback (https://www.ecil-leukaemia.com/en/resources/resources-ecil).

Search strategy and selection criteria

A systematic review was conducted using the Ovid platform of PUBMED to identify relevant English-written publications between 01/01/2010 and 15/09/2024. The searches included a combination of indexed terms and free-text terms ("haematology", "leukaemia", "stem cell transplantation", "CAR-T", "lymphoma", "Clostridioides", "Clostridium", "prophylaxis", "treatment", "prevention", "relapse", "recurrence"). Previous randomised controlled trials and meta-analyses were also reviewed (see Supplementary Material for details).

Role of funding source

None of pharmaceutical companies which supported ECIL with unrestricted grants had any role in selecting experts, determining the scope and purpose of the guidelines, collecting, analyzing, and interpreting the data, or preparing the guidelines' edition.

Results

Epidemiology of CDI in patients with haematological malignancies and after cellular therapies

Incidence

In twenty-nine identified papers (Table S1), the incidence of CDI was the highest in patients with acute leukaemia (5.2–11.8%), and the lowest in patients with

	Significance
Strength of a recommendation (SoR)	
Grade A	ECIL strongly supports the recommendation for use
Grade B	ECIL moderately supports the recommendation for use
Grade C	ECIL marginally supports the recommendation for use
Grade D	ECIL supports a recommendation against use
Quality of evidence (QoE)	
Level I	Evidence from at least one properly designed, randomised, controlled trial (orientated on the primary endpoint of the trial)
Level II	Evidence from at least 1 well-designed clinical trial (including secondary endpoints), without randomisation; from cohort or case-controlled analytic studies (preferably from >1 centre), from multiple time series, or from results of uncontrolled experiments **Added index for source of level II evidence** r: meta-analysis or systematic review of randomised controlled trials t: transferred evidence, that is, results from different patient cohorts, or a similar immune status situation h: comparator group is a historical control u: uncontrolled trials a: published abstract (presented at an international symposium or meeting)
Level III	Evidence from opinions of respected authorities, based on clinical experience, descriptive case studies, or reports of expert committees

Table 1: **Evidence-based medicine grading system established by the European Society for Clinical Microbiology and Infectious Diseases (ESCMID), the European Confederation of Medical Mycology (ECMM), and the European Respiratory Society (ERS).**[18,19]

chronic lymphoproliferative disorders (1%–4.8%).[3,20–23] The rate of severe and complicated CDI was 3–7% but definitions used varied. The rate of rCDI depended on time of evaluation of recurrence, and ranged from 4.6% to 27%, the latter in patients treated mainly with metronidazole.[3,21]

In HCT recipients, the incidence of CDI increased overtime: from 7.9% in the meta-analysis from 2014 (higher after allogeneic HCT: 9–34%, compared to autologous HCT (AHCT): 5–24%), to 13.2% in the meta-analysis from 2022 (15.3% after allogeneic HCT and 9.2% after AHCT).[24,25] Most CDI cases were diagnosed in the early post-transplant period (median 8–13 days after HCT).[26,27] Considering high rate of diarrhoea early after HCT, and the limitation of molecular methods, the CDI incidence might be overestimated.[14]

After CAR-T therapy the incidence varied significantly from 0% to 15.3%, with the median rate of 7.3% in recent studies, and occurred mainly early after CAR-T.[28] High rate of CDI reported in some studies, and the results of a meta-analysis showing that almost half of reported microbiologically documented infections of bacterial origin were CDI, highlight the need for antimicrobial stewardship in CAR-T setting.[29]

Risk factors for CDI

Identified risk factors in non-HCT setting included prolonged hospitalizations, mucosal damage and disruption of microbiota by conditioning regimens, acute leukaemia, prior colonization with *C. difficile* and broad-spectrum antibiotics.[3,20–22] In HCT setting, the risk factors were allogeneic HCT, male gender, leukaemia, previous antibiotic use and mechanical ventilation, prior hospitalization or chemotherapy, myeloablative conditioning regimens or total body irradiation, GvHD, particularly gastrointestinal, previous reactivation of CMV or other Herpesviridae, and, most importantly, prior colonization (see Supplementary Material).

Clinical impact of CDI

CDI in HM and HCT patients was associated with increased mortality in some studies, but not others (Tables S1 and S2). CDI-related mortality rates in HM and HCT patients were low, at 0–1% and 3%, respectively.[3,22,25] In a large US study, allogeneic HCT recipients with CDI had 3.7-fold higher mortality when compared to non-transplant patients, and in adjusted analysis, CDI was a significant predictor of mortality in patients with GvHD (OR 1.55).[30]

CDI also prolonged hospital stays and increased the risk of nosocomial infections and bloodstream infections (BSIs) in HM/HCT setting, further complicating patient management and possibly contributing to increased mortality.[31] In patients with CDI post-HCT, up to 20.3% developed severe disease (different criteria were used), 5% were ICU-admitted, 2.5% developed pseudo-membranous colitis, 1.3% perforation and 1.7% underwent colectomy.[25,31]

The bi-directional association and overlap between gastrointestinal GvHD (GI-GvHD) and CDI, with CDI increasing the risk of GvHD, and GvHD increasing the risk of CDI, adds complexity to the diagnosis and treatment of CDI after allogeneic HCT.[26,27,32] In a study of 826 patients with CDI, CDI was associated with lower overall survival and 2.58-fold increase in infection-related mortality (IRM), but when acute GvHD coexisted with CDI, IRM increased fourfold.[27]

Co-infections and differential diagnosis in Clostridioides difficile *infection (CDI)*

Differential diagnosis of diarrhoea in HM patients include both other infections and non-infectious causes. Diagnostic testing in case of diarrheoa usually includes both CDI and other gastro-intestinal pathogens.[26] Limited number of available studies reported low rate of coinfections with viral (including <1% rate of gastrointestinal CMV disease) or bacterial pathogens such as *Salmonella, Shigella, Yersinia, Campylobacter*.[27,33–36] However, this might differ if molecular panels are used, as reported in the general population with CDI, in which co-pathogens, mainly different strains of *Escherichia coli*, were very frequent.[37] Clinical significance of molecular detection of these pathogens, which might be colonisers in some patients, needs to be carefully evaluated. Indeed, in the settings of HCT and acute leukemia, molecular gastrointestinal panels were found of limited benefit in the differential diagnosis of cases of nosocomial diarrhoea due to high sensitivity but without distinguishing between colonisers (especially toxin-negative CD and diarrheagenic gram-negative organisms) and pathogens being true causes of diarrhea, while they were much more useful in cases of community-acquired diarrhoea.[38,39] Nevertheless, in an analysis of data from a RCT on CDI, lower rates of sustained clinical cure were reported in case of detection of CDI and co-pathogens, mainly enteropathogenic and Shiga toxin-producing *E. coli*.[40] Therefore, testing for other infectious causes is fundamental in case of CDI treatment failure.

Essential non-CDI diagnostic work-up should include main viruses (i.e. norovirus, rotavirus, adenovirus), enteropathogenic bacteria (i.e. *Salmonella, Campylobacter, Shigella,* diarrheagenic strains of *E. coli)*, and, particularly in endemic regions, protozoa such as *Giardia lamblia, Cryptosporidium, Entamoeba histolytica*, and *Cyclospora cayetanensis*, using best available microbiological testing, such as microscopy, culture and multiplex PCR, especially for difficult to culture pathogens. The advantages of molecular panels are high sensitivity, concurrent testing for multiple pathogens, and rapid time to results, while their limitations include possible false positive results due to cross-reactivity, detection of pathogens that might not be responsible

for the episode of diarrhoea (e.g. enteropathogenic *E. coli*), and failure to detect some pathogens, depending on the panel performed. Testing for CMV-DNA should be performed only in patients with clinical suspicion of CMV disease, since fecal shedding can occur and diagnosis of CMV disease requires further confirmation.[41]

Non-infectious causes of diarrhoea are frequent in patients with HM/HCT, and include chemotherapy, conditioning regimen, mucositis, gastrointestinal GvHD, or immunosuppressive drugs. Indeed, most of HCT recipients diagnosed with CDI had also other potential causes of diarrhoea such as mucositis or drugs used in conditioning.[15] Typical non-infectious causes of diarrhoea in HM/HCT patients will vary based on the underlying disease and its treatment, but causes not specific to immunocompromised population, such as enteral feeding or ischemia, should also be considered. Mucositis and chemotherapeutic drugs such as anthracyclines, or new agents such as midostaurin, are frequent causes of diarrhea in case of chemotherapy for acute myeloid leukemia. In HCT setting, mucositis, conditioning regimen related toxicity, cyclophosphamide, or engraftment syndrome are frequent causes of diarrhoea during pre-engraftment phase, while gastrointestinal GvHD, transplant-associated thrombotic microangiopathy, and diarrhoea due to drugs such as mycophenolate mofetil or checkpoint inhibitors occur during post engraftment period.[42] After CAR-T treatment, diarrhoea can occur during cytokine release syndrome, or much later due to CAR-T specific immune effector cell-associated enterocolitis, and clinical presentation with pseudomembranes has been reported.[43,44] In chronic hematological malignancies, treatment with proteasome inhibitors, idelalisib, or lenalidomide might be a cause of mainly mild but chronic diarrhoea.[45]

At the onset of diarrhoea initial diagnostic work up should exclude main infectious causes, while comprehensive diagnostic work-up is essential in patients who do not respond to standard CDI therapy after 3 days of treatment, or have only partial response despite full treatment course, particularly when isolated NAAT positivity is present without corresponding toxin detection.

Paediatric patients

Irrespective of the fact that asymptomatic colonization rates of 30–50% have been reported, *C. difficile* is a common cause of infectious diarrhoea in paediatric patients with cancer and HCT, with CDI rates between 5 and 30%. Risk factors for CDI are similar to those observed in adults and include prolonged hospitalization, exposure to antibacterial and cytostatic agents, and use of proton pump inhibitors. Using adult definitions, severe cases are reported in 0–20% of episodes, and attributed mortality ranges from 0 to 3.4%.[46–49] In a mixed cancer/HCT population, enteric co-infections have been reported in 19% of CDI cases. Recurrent CDI may occur in up to 27% of patients; however, limited data exist on rCDI risk factors.[46–50]

Diagnosis of CDI

Diagnosing CDI in HM/HCT patients is complicated by high rates of asymptomatic colonization with toxigenic *C. difficile* strains and the presence of non-infectious causes of diarrhoea. The overview of diagnostic tests is provided in Table S3. Diagnostic tests should be performed only in symptomatic patients. Clinical judgment is crucial in testing and result interpretation, particularly if only molecular testing is available, or if toxin detection is negative and only molecular test is positive, since there is risk of over-diagnosis and treatment of patients colonised with toxigenic *C. difficile* who have diarrhoea due to a different cause.[13,15] In such cases, careful evaluation of other causes of diarrhoea, particularly in case of lack of response to CDI treatment is mandatory. The recommendations for the diagnosis of CDI, including annotations for children, are provided in Table 2.

Definitions

Consensus definitions were formulated to advance the standardized interpretation and reporting of CDI in HM/HCT patients (Table 3). Recurrence should be defined by the new onset of compatible symptoms following clinical response and completion of therapy, with new laboratory test confirmation. The timeframe for defining an episode of recurrence remains challenging with various timepoints used, with most studies considering 60 days (Tables S1 and S2). For CDI treatment trials, a fixed period from 4 to 12 weeks from the end of therapy was used, while guidelines recommended 8 weeks.[54–56] However, we concluded that a period of 12 weeks following initial CDI episode is more reflective of clinical practice and relevance in the setting of repeat or prolonged hospital admissions. In HM/HCT, clinical failure should be diagnosed in case of progression to severe or complicated CDI, while in case of persistent symptoms, other causes of diarrhoea should be searched for.

To advance uniform reporting of CDI in clinical studies, criteria for defining diagnostic probability of CDI were proposed (Table 4). Level of certainty is reflected through the use of possible, probable and proven categories and requires assessment of 3 domains: 1) clinical with compatible signs and symptoms, 2) laboratory testing for CDI with increasing diagnostic specificity in case of positive toxin result, and 3) likelihood of other potential aetiology of diarrhoea based on both clinical and laboratory exclusion (Fig. 1). These criteria are not intended for purposes of clinical management, in particular treatment initiation.

	Grading
Recommendation	
Test for CDI only in case of compatible signs and symptoms	A II u
Screening for asymptomatic colonization is not recommended	D II u
For stool testing use diagnostic algorithms that include both toxin detection with immunoenzymatic assay (more specific for disease) and a more sensitive test for detection of toxigenic CD in accordance with ESCMID recommendations: (1) either toxin plus NAAT or (2) Glutamate Dehydrogenase (GDH) plus toxin and, if GDH positive and toxin negative, confirm with NAAT. Rectal swabs can be tested with NAAT in case of ileus	A II t u
In case of partial or absent clinical response to CDI treatment, additional testing for other pathogens should be performed and non-infectious causes of diarrhoea should be considered (see text for details)	A II t
No repeated testing is necessary in case of a negative result obtained during the previous 7 days of the same diarrhoea episode	B II t
Testing for proof of cure is not recommended	D II t
Comments on testing in children	
• Testing for *C. difficile* should be performed in immunocompromised children ≥2 years of age that are symptomatic and fulfil the criteria for diarrhoea • In the population aged <2 years, high rates of asymptomatic colonization should be taken into account, and testing is recommended only in case of high clinical suspicion[a] • The algorithm for testing in children remains the same as per the adult recommendation • Laboratory diagnostics for CDI should be combined with diagnostics for other gastrointestinal pathogens relevant in the immunocompromised paediatric host • Clinical correlation should guide the interpretation of laboratory results and the decision to treat	

CDI, *Clostridioides difficile* infection; ESCMID, The European Society of Clinical Microbiology and Infectious Diseases; GDH, Glutamate Dehydrogenase; NAAT, Nucleic Acid Amplification Test. [a]Infants are frequently colonized with both toxigenic and nontoxigenic *Clostridioides difficile*: Colonization rates in otherwise healthy infants range from 40 to 80% under 1 month, 30% under 6 months, and 14% between 6 and 12 months of age. The frequency remains as high as 22% in toddlers 1–2 years of age before declining to approach rates seen in healthy adults of 1%–3%,[48,51] while colonisation with toxigenic strains of 13–14% between the age of 6 and 24 months.[52] Asymptomatic colonization in this period of life may hypothetically be due to the relative absence of cellular pathways necessary for pathogenicity (toxin receptors or downstream signalling pathways) in the immature gut mucosa. The spontaneous eradication after the first year of life is likely to result from ecological competition with the adult-type microbiome, which suppresses the growth of *C. difficile* (ecological succession).[53]

***Table 2*: ECIL-10 Recommendations on diagnosis of CDI in adult and paediatric patients.**

Defining severe CDI

Severity scoring for CDI in the general population has never been validated in HM/HCT population and can be of limited utility in this population. Classical severity indicators, such as leukocytosis, can be unreliable in this setting due to influence of underlying malignancies and its treatments.[57] Moreover, the clinical presentation of CDI may be muted by neutropenia and immunosuppression, reducing the frequency of inflammation-mediated complications, such as pseudomembranes.[58–61] Neutropenia cannot be considered a definitive indicator of CDI severity since colonization often occurred during neutropenic periods.[13–15,21,34,62–65] Although severe abdominal pain requiring analgesia has been considered as a marker of severe CDI,[66,67] it is nonspecific and insufficient as an independent severity criterion.[68] Observational studies reported low rates of classic severity indicators, such as leukocytosis, hypoalbuminemia or acute renal failure,[2,57,64,69–75] The presence of non-gastrointestinal co-infections, such as CMV reactivation or BSIs, has been reported and may hamper correct evaluation of CDI severity in this population.[74]

The goal of managing CDI in HM/HCT is early identification of individuals at high risk of complications, rCDI, or death, to ensure timely and appropriate interventions. Hematologic malignancy alone should not be regarded as a severity criterion, nor it should be viewed as a direct risk factor for complicated CDI.[2,69] Similarly, neutropenia is not a definitive indicator of CDI severity.

The only study reporting severity criteria specific to allogeneic HCT patients to be associated with mortality, defined CDI severity based on the volume of diarrhoea present within 48 h from CDI diagnosis.[76] Mild CDI was characterized by grade 1 diarrhoea (≤500 mL/24 h) and/or colitis, moderate by grade 2 (501–1000 mL/24 h), and severe by grade 3 or higher (≥1001 mL/24 h) and/or colitis.[76] Median survival rate after CDI was higher in mild or moderate compared to severe cases, altough study limitations should be acknowledged.[76] An overlap between severe CDI and severe GI-GvHD may occur, as both conditions use diarrhoea volume to estimate severity. This overlap may influence the evaluation of the impact of severe CDI on survival.

The criteria to define severe and severe-complicated (including fulminant CDI) are provided in Table 5. While ideal definitions for paediatric patients do not exist, definitions for severe CDI are mostly congruent and those for severe-complicated CDI similar to adult population.

Treatment of CDI

All treatment recommendations are summarised in Table 6.

Treatment of primary episode of CDI, non-severe and severe

Recommendations on antibiotic treatment of the first episode of CDI are based on data largely extrapolated from trials conducted in the general population and smaller sub-groups of immunocompromised or cancer patients. Fidaxomicin had similar rates of clinical cure

Clinical picture compatible with CDI	• Presence of diarrhoea (≥3 unformed stools/24 h), in the absence of the use of laxatives OR • In patients with pre-existing diarrhoea or altered bowel habits (e.g. in case of ostomy) – significant worsening (particularly if with additional negative impact on daily activities) OR • Ileus or megacolon Additional signs and symptoms suggestive of severe CDI: pseudo-membranes, colonic thickening in the absence of other causes of colitis Exclusion of other potential causes should be pursuit following local protocols
Colonisation	Absence of symptoms/signs but laboratory detection of toxigenic *C. difficile*
Clinical response (final evaluation 48 h after end of treatment)	Resolution of symptoms OR Sustained improvement (<3 unformed stools/day) for at least 48 h during therapy OR Significant reduction in unformed stools/day in case of pre-existing diarrhoea OR Attainment of bowel movements of Bristol Stool Form Scale types 1–4
Recurrence	Clinical • New onset of symptoms (as per initial episode) after clinical response • Following completion of therapy With laboratory confirmation • Positive laboratory test as per initial episode Timeframe: In the literature multiple time frames have been utilised, either with diagnosis of CDI or end of treatment as starting point. For CDI treatment trials: 30 days or 8 weeks from the end of treatment For clinical studies/practice: 12 weeks from diagnosis of CDI (with both clinical and laboratory criteria)

***Table 3:* Definitions related to *Clostridioides difficile* infections in adult and paediatric patients.**

at the end of treatment compared to vancomycin (88–100% versus 80–86%) but lower rates of recurrence by the end of study (13–15% versus 15–27%), resulting in higher rate of global clinical cure with fidaxomicin (75–80% versus 62–64%).[55,77–79] Observational studies in HM/HCT patients noted similar findings.[80–82] In post-hoc analyses, rates of recurrence were significantly lower with fidaxomicin in patients with cancer and in those receiving concurrent antibiotic therapy.[83,84] Lower rates of recurrence were also reported with the use of extended-pulse dosing of fidaxomicin compared to standard course of vancomycin, which can be particularly useful in selected situations, such as concomitant antibiotic therapy.[40,85] Fidaxomicin has less impact on the microbiome.[86,87] Therefore, fidaxomicin is preferred in patients at high risk of rCDI or microbiota disruption-related complications such as severe CDI, concomitant antibiotic treatment, and its main drawback is high acquisition cost.

In contrast, clinical trials and meta-analyses demonstrated significantly lower rates of clinical cure, sustained cure and higher recurrence rates with metronidazole compared to vancomycin or fidaxomicin, especially in immunocompromised patients.[88–93] These findings were also reinforced in observational studies.[26,94,95]

In case when oral intake is impossible, oral administration of first line agents is recommended through nasogastric tube (NGT). High dose of oral vancomycin (500 mg QID) was found of no benefit over standard dose (125 mg QID) in non-fulminant CDI.[96] Despite lack of comparative data, high dose has been consistently used in case of NGT administration in most studies. Crushed fidaxomicin is stable and can be administered through NGT.[97,98] Intravenous metronidazole has lower efficacy, and this should be taken into consideration when balancing the feasibility of oral intake and the need for NGT placement, particularly in patients with severe mucositis and thrombocytopenia.[99]

Oral teicoplanin was approved for CDI treatment, based on efficacy reported in 2 small RCTs in '90, but is not universally available.[90]

Resistance of *C. difficile* to vancomycin and metronidazole is very low (<1–3%), and identification of

Level of certainty	Clinical	Laboratory	Other causes of diarrhoea
Proven	Presence of diarrhoea (≥3 episodes of unformed stools/24 h) or significant worsening of pre-existing diarrhoea or ileus/megacolon	Direct detection of toxin of *C. difficile*	Non-infectious causes considered to be unlikely on clinical assessment[a]
	Histopathological confirmation of pseudomembranes	Detection of toxigenic *C. difficile*	No other infectious causes Formal microbiological exclusion of co-infections is recommended for clinical trials
Probable	Presence of diarrhoea (≥3 unformed stools/24 h) or significant worsening of pre-existing diarrhoea or ileus/megacolon	Direct detection of toxin of *C. difficile*	Other non-infectious or infectious causes possible
		Detection of toxigenic *C. difficile*	No other non-infectious and other infectious causes
Possible	Presence of diarrhoea (≥3 unformed stools/24 h) or significant worsening of pre-existing diarrhoea or ileus/megacolon	Detection of toxigenic *C. difficile*	Other non-infectious or infectious causes possible

[a]Consideration of non-infectious causes: GvHD, mucositis, chemotherapy or conditioning regimen, certain drugs.

***Table 4:* Proposed criteria for defining different levels of probability/certainty of CDI diagnosis in adult and paediatric patients, developed for clinical studies and not for individual treatment decisions.**

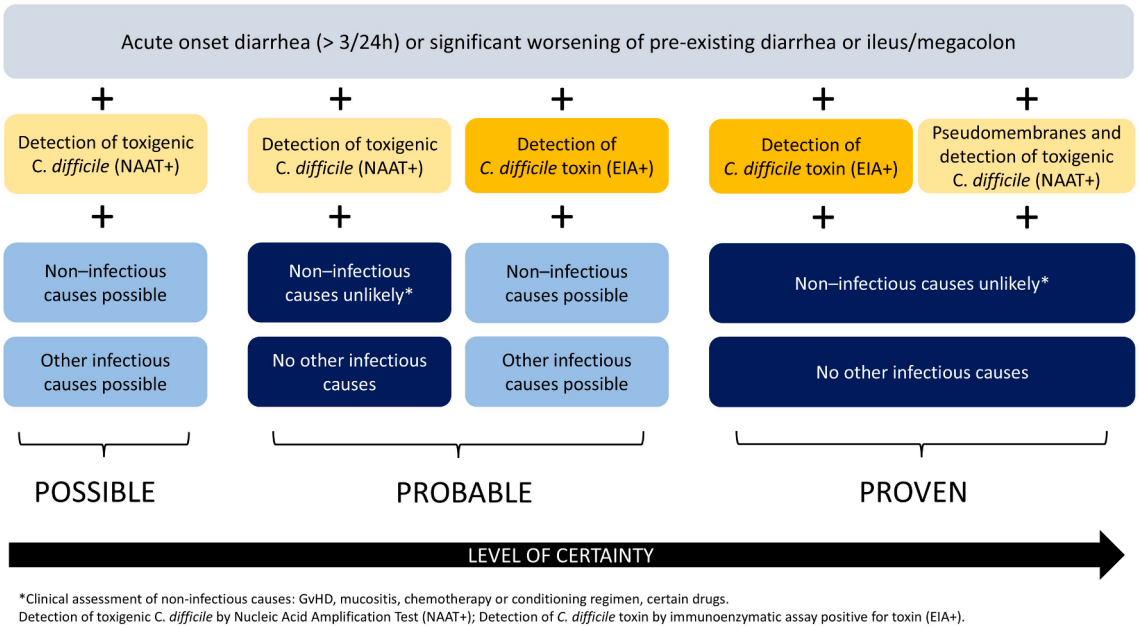


Fig. 1: Defining levels of certainty in the diagnosis of CDI in patients with haematological disorders and undergoing cellular therapy.

mutations that confer resistance to fidaxomicin is very rare.[100,101]

Additional considerations and adjunctive measures to treatment

In case of CDI, discontinuation of other concomitant antibiotics should occur when feasible.

Use of bezlotoxumab, a monoclonal antibody against *C. difficile* toxin B, in addition to CDI-directed antibiotic therapy resulted in significantly lower rates of rCDI within 12 weeks (17% versus 27%), largely in hospitalised patients with CDI or rCDI.[54] Two post-hoc analyses demonstrated similar findings for immunocompromised or cancer patients.[4,5] Its benefit was derived mostly in combination with either metronidazole or vancomycin (97%) and rates of recurrence with placebo were higher (27–30%) than commonly observed.[4,25,27,54] A slightly increased risk of heart failure, particularly in patients with pre-existing cardiac conditions was reported.[54] However, its production was discontinued in 2025.

Treatment of severe-complicated, including fulminant, CDI

Patients with severe CDI were included in randomized controlled trials (22%–50% of included patients) with similar efficacy reported for fidaxomicin versus vancomycin.[55,79,89] However, patients with severe-complicated or fulminant CDI were excluded from all RCTs. Therefore, data on their treatment came from observational, mainly retrospective studies.

With complicated/fulminant CDI, oral vancomycin plus intravenous metronidazole is considered standard of care and recommended by most guidelines but supporting data is limited to few studies comparing addition of IV metronidazole to vancomycin versus vancomycin monotherapy.[95,102–105] These studies have the intrinsic bias of providing more aggressive treatment to patients in poorer clinical conditions. Consequently, outcomes were usually worse in combination treatment subgroups.[103–105] Only in one study (with limitations) the advantage of combination therapy on mortality was reported.[106]

Tigecycline has high faecal concentrations and low MIC values in *C. difficile*, but limited data are available on its use for CDI.[107–109] One retrospective study on tigecycline monotherapy found higher rates of clinical cure compared to the combination of vancomycin and metronidazole.[107] A review of retrospective studies showed clinical cure of 79%, but tigecycline was used mostly in combination with other agents.[108] In a propensity-matched retrospective study, addition of tigecycline did not improve 30-day mortality.[109]

Similarly to what is recommended for severe CDI, in case when oral intake is impossible, oral administration of first line agents is recommended through NGT. There is limited clinical data to support use of fidaxomicin in complicated/fulminant CDI.[110]

In case of ileus or megacolon, endorectal vancomycin 500 mg every 6 h should be added to the oral and intravenous treatment of severe CDI to achieve adequate colonic concentrations.[111] It is recommended by most guidelines, although the quality of evidence is low.

Use of IVIg in fulminant CDI has been proposed with the aim of reducing the severity also due to toxin

Severe CDI: ≥1 of the following criteria	Severe-complicated CDI: ≥1 of the following criteria
Diarrhoea >1000 mL or ≥6 unformed (i.e. type >4 on Bristol Stool Form scale) stools/24 h[a]	Hypotension or septic shock
Rise in creatinine >50% baseline[b]	ICU admission for CDI
Leukocyte count >15 G/L[c]	Including **fulminant CDI**, defined as: ileus, megacolon, perforation, abdominal surgery for CDI
Colitis including distension of large intestine, pericolonic fat stranding, colonic wall thickening, pneumatosis intestinalis	
Pseudomembranous colitis	

Of note, ideal definitions for severe or complicated CDI in paediatric cancer/HCT patients do not exist. Proposed clinical and radiographic criteria include: at least ten stools per day; hypotension requiring intervention; ventilation support (invasive and non-invasive ventilation); total parenteral nutrition; nil per os due to gastrointestinal signs or symptoms; abdominal pain requiring opioids; abdominal imaging abnormalities including colitis, typhlitis, enteritis or pneumatosis intestinalis; clinical diagnosis of ileus; pseudomembranous colitis upon endoscopy; intensive care unit admission; surgery for CDI. [a]Not applicable if pre-existing diarrhoea. [b]Without another cause of renal failure, e.g. drug toxicity. [c]Except in cases in which leucocyte count is not interpretable (neutropenia, ongoing or recent chemotherapy, GvHD, recent cellular therapy, hyperleukocytosis for other reason, etc.).

Table 5: **ECIL-10 definition for severe or complicated CDI.**

binding before bezlotoxumab data were made available. Data are too limited to provide recommendations, as they come mainly from case reports and very heterogenous case series.[112,113]

Multidisciplinary consultation with a surgeon and an ID specialist is recommended. Surgical considerations include indication for diverting loop ileostomy (DLI), with the aim of avoiding colectomy, based on data from observational studies including immunocompromised patients reporting similar or lower mortality, and a possibility of colon preservation with DLI.[114,115]

There is no standard duration of treatment of complicated/fulminant CDI and additional secondary preventive strategies can be applied if appropriate (tapered-pulse treatment, bezlotoxumab, FMT).[116]

Treatment of recurrent CDI (second and subsequent episodes of CDI), including in the setting of concurrent antibiotics which cannot be discontinued

Prevention of rCDI in HM/HCT population is a major challenge, due to co-existence of well recognized risk factors for rCDI, such as prolonged or repeated antibiotic exposure, prolonged hospital admission and chemotherapy-induced microbiota disruption.

Secondary prevention strategies include: 1) use of treatment regimens associated with lower recurrence rates, 2) tapered (i.e. progressively reducing the daily dose) and/or pulsed (i.e. administration on non-consecutive days) administration schedules of vancomycin or fidaxomicin, 3) prolonging treatment duration, e.g. in case of concomitant antibiotic treatment, 4) administering pharmacological secondary prophylaxis during high-risk periods, 5) use of adjunctive therapies such as bezlotoxumab, FMT or live biotherapeutic products.

Patients with prior CDI constituted 17–22% of trial populations in which the use of fidaxomicin (standard or extended-pulse dosing) resulted in significantly lower rCDI.[40,55,79,84] In the extended-pulse trial of fidaxomicin, approximately 70% of patients were receiving concomitant antibiotics, and lower recurrence rate was observed compared to standard 10-day vancomycin treatment.[40] Other studies have reported lower rCDI rates with taper-pulse use of vancomycin over several weeks.[117–120] Currently there are no published studies directly comparing extended-pulse dosing of fidaxomicin with taper-pulse dosing of vancomycin.

For patients at highest risk for rCDI, the addition of bezlotoxumab could be considered in line with trial and post-hoc analyses results.[4,5,54]

Monotherapy with metronidazole was associated with lower rates of sustained clinical cure and higher rates of recurrence and therefore is not recommended for treatment of rCDI.[88–90]

Patients with HM/HCT were frequently treated with longer duration of CDI therapy (mostly metronidazole) with a median duration of 16 days.[34,121,122] Indirect data from observational studies suggested that prolonging CDI-directed antibiotics until conclusion of concurrent antibiotic therapy was associated with lower rates of subsequent recurrence.[6,22,123] In a small randomised trial in patients treated with concomitant antibiotic therapy, fidaxomicin and vancomycin administered until systemic antibiotic discontinuation were associated with similar rates of clinical cure and rCDI.[124] Maintenance dose of vancomycin 125 mg BD post-initial treatment of CDI with vancomycin has been associated with lower rates of rCDI and could be a viable strategy.[123]

For secondary prophylaxis during at-risk periods following conclusion of primary treatment, FMT or other live microbiota interventions, see dedicated paragraphs.

Recommendations for the treatment of rCDI are summarised in Table 6.

Secondary prophylaxis with CDI-directed antibiotics

Secondary prophylaxis with oral vancomycin, rifaximin, or fidaxomicin during periods of high risk, such as ongoing antibiotic therapy, has been explored in

Intervention	Grading	Comment
Primary episode, non-severe and severe		
Discontinuation of other concomitant antibiotics should occur when feasible	Good clinical practice statement	
Fidaxomicin 200 mg BD	A II t u r	Fidaxomicin and vancomycin were associated with similar rates of cure at the end of treatment Fidaxomicin was associated with lower rate of recurrence and less impact on microbiome and VRE selection Thus, fidaxomicin should be preferred in patients at high risk of rCDI or microbiota disruption-related complications, e.g. severe CDI, concomitant antibiotic treatment, allogeneic HCT
Vancomycin 125 mg four times a day, oral solution or tablets	A II t u r	
Metronidazole monotherapy, IV or oral, is not recommended as the first line treatment for primary episode of CDI	D II t	No longer recommended as first line due to lower rates of cure at end of treatment and higher rates of recurrence[a]
If oral intake is not possible, nasogastric tube (NGT) administration of first line treatment options with or without concurrent administration of intravenous metronidazole is recommended[a]	B II t	
The addition of bezlotoxumab to CDI antibiotic therapy can be considered in patients at highest risk for rCDI	B II t	Evidence of benefit mainly if CDI therapy with vancomycin or metronidazole. Bezlotoxumab is currently out of commercial production
Duration		
Duration of 10 days of therapy is recommended	A II t	
For fidaxomicin, extended pulse dosing with the same number of tablets could be utilised	B II t	Utilize the same total number of tablets over 25 days versus 10 days to reduce the rate of recurrence (200 mg BID for 5 days, then 200 mg every 48 h), particularly useful e.g. in patients on concomitant antibiotics
Severe complicated/fulminant CDI		
Vancomycin, oral or via gastroenteric administration through NGT plus either	A II t	Most observational data with metronidazole plus vancomycin Limited published data on tigecycline, mainly in-vitro, observational studies used in complicated infection, mainly in combination
Metronidazole IV 500 mg three times a day OR	A II u	
Tigecycline IV standard dose	C II t r	
In case of ileus and/or megacolon		
Add endorectal vancomycin 500 mg every 6 h to the regimen recommended for severe complicated/fulminant CDI	A II t	
Multidisciplinary consultation with a surgeon, also to consider diverting loop ileostomy (DLI) aimed to avoid colectomy, and an ID specialist is recommended	B II t r	Mortality similar to colectomy, but colon preservation with LI, no specific data in immunocompromised
Recurrent CDI episode (second or greater CDI episode)		
Fidaxomicin 200 mg BID	A II t	Particularly if not used during the previous episode(s) Standard 10 days course or the extended-pulse dosing can be used
An alternative treatment is vancomycin 125 mg four times a day for 10 days, then taper-and-pulse regimen over 4 weeks	A II t r	Taper and pulse regimen: 125 mg three times a day for 1 week, then 125 mg twice a day for 1 week, then 125 mg every 24 h for 1 week, then 125 mg every 48 h for 1 week No data on treatment of r-CDI in HM/HCT
Vancomycin standard course could also be considered	B II t	
Metronidazole monotherapy IV or oral	D II t	Not recommended due to lower efficacy
The addition of bezlotoxumab to CDI antibiotic therapy can be considered in patients at highest risk for recurrent CDI	A II t	To be considered in patients at highest risk of CDI recurrence Evidence of benefit in addition to standard of care antibiotics, mainly if CDI therapy with vancomycin or metronidazole Bezlotoxumab is currently out of commercial production
Recurrent CDI episode (second or greater CDI episode) in case of concomitant antibiotic therapy which cannot be discontinued during treatment of the CDI episode		
Fidaxomicin, with a preference of extended pulse dosing	B II t	
Vancomycin can be utilised with alternative dosing strategies such as taper-pulse regimen or low dose maintenance therapy following standard dosing	B II t u	Indirect evidence suggests that continuation of CDI treatment until completion of concurrent antibiotics could be associated with lower rates of recurrence
Use of feacal microbiota transplantation (FMT) and live biotherapeutic products (LBP) in HM/HCT		
There are no data to routinely recommend FMT in haematological patients for the treatment of rCDI		
After multidisciplinary consult, FMT can be considered for multiple recurrent CDI in non-profoundly neutropenic patients	B II t	
More data are required in profoundly neutropenic and GvHD patients and in fulminant CDI		

CDI, *C. difficile* infection; FMT, faecal microbiota transplantation; GVHD, graft versus host disease; NGT, nasogastric tube; rCDI, recurrent CDI. Paediatric approval status and age-specific dosing recommendations can be found it Table S7 of the supplemental materials. [a]Metronidazole iv monotherapy might be reasonable for non-severe CDI in selected paediatric patients with inability of oral intake and no NGT in place, balancing the lower efficacy against the need for NGT placement.

Table 6: **ECIL-10 recommendations for adult and paediatric patients on treatment of CDI, divided into: primary episode (non-severe and severe), severe complicated or fulminant CDI, recurrent CDI episode (second or greater), recurrent CDI episode (second or greater) in case of concomitant antibiotic therapy which cannot be discontinued during treatment of the CDI episode.**

12 studies (see Table S4). Vancomycin (various doses) has been evaluated retrospectively with mixed results ranging from no benefit to lower likelihood of rCDI with oral vancomycin 125 mg BD until 7 days post antibiotic discontinuation. Infection or colonization with VRE was not different with the use of vancomycin prophylaxis. Two small prospective randomised studies, performed in general population, reported a trend for benefit of rifaximin while a small retrospective study found no benefit, with rCDI rate of 36%.

Thus, secondary prophylaxis with vancomycin is recommended only in selected patient populations until discontinuation of antibiotics (Table 7).

Faecal Microbiota Transplantation (FMT)

FMT was reported efficacious (83–92%), safe and cost-effective in clinical trials with mixed-patient population of largely non-immunosuppressed patients, but patients with severe disease or prior immunosuppression were excluded from RCT.[125]

A growing body of evidence supports FMT's efficacy and safety in patients with moderate immunosuppression, and its use in HM and HCT patients has been described as effective, particularly in those who are not profoundly neutropenic. FMT for CDI has been reported as mostly successful in approximately 70 patients with HM, but only 2 were severely neutropenic (Table S5).

Safety data can be also extrapolated from the use of FMT in treatment of GI-GvHD in more than 100 patients, with some reports of severe BSIs (Table S5). However, BSI is a frequent complication in severe GvHD, thus the role of FMT remains unclear.

A European Bone Marrow Transplant (EBMT) survey found that 77.4% of respondents considered FMT safe in allogeneic HCT patients, although only a minority of adult and paediatric centres had adopted its use.[126]

As efficacy might be compromised in case of the need for concomitant antibiotic treatment, repeated administration of FMT might be required. Further studies are needed to clarify its efficacy and role in profoundly immunocompromised patients, including those with GvHD and fulminant CDI.

Live biotherapeutic products

Live biotherapeutic products (LBP) were developed for prevention of rCDI by reducing the disruption of microbiome. Compared to FMT, these products are more standardized, readily available on a large scale and easy to administer. They can either originate directly from human donor stool or are designed microbial consortia manufactured *ex vivo*, and are administered either as enema or orally. Two products (donor-derived enema and donor-derived capsules) have been FDA approved, but immunocompromised patients were not included in the trials.[127,128] Thus, the role LBP in HM/HCT setting remains to be determined.

Further data are keenly awaited to confirm if FMT and LBP could restore intestinal microbiota, and hopefully reduce the risk GI GvHD, reduce the rate of colonisation with resistant pathogens and improve patients' survival.

Primary prophylaxis of CDI

There are limited data on primary CDI prophylaxis in high-risk patients (see Table S6). In four studies (including one RCT) oral vancomycin (125 mg QD or BD) reduced the incidence of CDI, with no difference in VRE colonisation or infection found in 2 studies. Three additional RCT assessed metronidazole, ribaxamase, and fidaxomicin. One was a randomised study of adult HCT recipients receiving fluoroquinolone prophylaxis which failed to demonstrate the pre-defined benefit of fidaxomicin 200 mg QD administered from conditioning until 7 days post-engraftment or until antibiotic prophylaxis discontinuation.[129] However, there were fewer cases of confirmed CDI in the prophylaxis arm.

These data suggest a potential benefit of primary CDI prophylaxis in high-risk patient populations, but data in HM patients remains limited. Considering the potential exposure of large number of patients to prevent one event of a relatively mild-to-moderate treatable infection, we do not recommend routine administration

Intervention	Grading
Secondary prophylaxis	
ECIL does not recommend routine administration of secondary CDI prophylaxis in haematology patients	D II
In selected patient populations at high-risk for recurrent CDI, such as patients with prior CDI and concomitant treatment with broad spectrum antibiotics, secondary prophylaxis with oral vancomycin 125 mg BD could be considered until discontinuation of antibiotic treatment	C II u
Primary prophylaxis	
ECIL does not recommend routine administration of primary CDI prophylaxis in haematology patients	D II
Fidaxomicin 200 mg QD or vancomycin 125 mg QD or BD along with strict infection control measures could be considered in very selected high-risk settings, such as centres with high prevalence during a CDI outbreak. In such cases, fidaxomicin is the preferred agent due to minor impact on gut microbiota	C II u
ECIL does not recommend probiotics as primary prophylaxis, for safety reasons during neutropenia, and due to the lack of demonstrated benefit in non-neutropenic/general population	D II t
There is not enough evidence to recommend vaccination for primary CDI prophylaxis in haematology patients.	

Table 7: **Recommendations for adult and paediatric patients on secondary and primary prophylaxis of CDI.**

of primary CDI prophylaxis in HM, except for special situations, such as centres with very high prevalence of CDI or in the setting of serious outbreak.

If primary CDI prophylaxis is considered, fidaxomicin or vancomycin can be used, with the former being preferred due to its minor impact on microbiota compared to vancomycin and because it is the only agent studied in a RCT in HCT recipients.[129]

Probiotics, though sometimes considered of benefit in patients receiving antibiotic therapy, are not recommended for primary prophylaxis in neutropenic patients or those with significant immunosuppression, as their safety (risk of bloodstream infections, e.g. due to *Saccharomyces* or *Lactobacillus*) and efficacy in these populations remain unproven. In non-neutropenic patients, their limitations are the lack of high-quality data proving their efficacy, limited standardization, and variation in composition.[130]

Five phase 2 or 3 RCTs on vaccination to prevent of CDI were published, but no data in HM/HCT patients are available. Thus, it cannot be recommended as a primary prophylaxis in this setting.

Paediatric considerations

Similar to previous ECIL guidelines and consistent with paediatric drug development regulations and guidelines from the European Medicines Agency (https://www.ema.europa.eu/en/documents/scientific-guideline/e-11-clinical-investigation-medicinal-products-paediatric-population-step-5_en.pdf; https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32006R1901), the recommendations for interventions were based on four essential components: evidence of efficacy from phase 2 and phase 3 trials corresponding to the adult recommendations, existence and quality of paediatric pharmacokinetic data and dosing recommendations; specific paediatric safety data and supportive efficacy data; and regulatory approval for use in paediatric patients by the European Medicines Agency.[131]

Based on the above criteria, the group had similar treatment recommendations for paediatric patients (Table 6). For paediatric approval status and age-specific dosing recommendations, please see Table S7 in supplemental materials.

Principles of CDI infection control

CDI in the hospital setting can be prevented through infection prevention practices and antibiotic stewardship programs. The cornerstones of CDI prevention do not differ from those in the general population and include: 1) early diagnosis; 2) immediate implementation of contact precautions and strict hand hygiene; 3) effective environmental cleaning to eliminate *C. difficile* spores; 4) identification and removal of environmental sources; 5) education and training; 6) surveillance and 7) antimicrobial stewardship to prevent unnecessary antibiotic therapies.

Infection control bundles for CDI prevention address key areas such as infrastructure, education, communication, and specific preventive measures, and should be implemented under the guidance of infection control and stewardship teams.

Detailed recommendations on CDI infection control are beyond the scope of these guidelines. It is recommended that current international infection control guidelines and national jurisdictional infection control policies be followed.[132–134]

Conclusion and outstanding questions

CDI continues to represent a significant clinical challenge in patients with HM/HCT, largely due to the lack of sensitive diagnostic tests that can distinguish true infection from colonization in patients who frequently experience diarrhoea from other non-infectious and infectious causes. Moreover, the prevention and treatment of rCDI remain difficult, given the frequent need for broad-spectrum antibiotic therapy for other infections, prolonged hospital stays, and profound microbiota disruption. Thus, treatments with the lowest negative impact on the microbiota should be favoured.

Future research should focus on the development and validation of new sensitive and specific diagnostic methods, including ultrasensitive toxin assays. The utility of new definitions for proven or probable CDI should also be evaluated. Tailored prevention and treatment strategies are necessary for this heterogenous and fragile population. Additionally, microbiota-based therapies should be investigated to demonstrate clinically meaningful and long-term benefits in patients with preexisting microbiome disturbances, such as HCT recipients.

Contributors

Conceptualization: MM, DN, AHG, BWT. Literature review and data curation MM, CR, DN, CO, AP, AHG, PM, BWT. Data analysis and creation of recommendations: MM, CR, DN, CO, AP, ER, AHG, BWT. Writing–original draft: MM, CR, DN, CO, AP, AHG, PM, BWT. Writing–review and Editing: all authors. All authors read and approved the final version of the manuscript. MM, CR, OC, AP, BWT, DN, MM and AG had access to and verified the underlying data.

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Data sharing statement

None.

Declaration of interests

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.eclinm.2025.103371.

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